

DE Metabolites Summary (v12 panel) - Detailed Pathway Grouped (with FDR)

Based on the new control set (result_v12 folder) using Standard Limma.
Colors: **red** = upregulated (logFC > 0), **blue** = downregulated (logFC < 0).

Comparison	P < 0.05	P < 0.01	FDR < 0.1	P < 0.01 Metabolites (Detailed Grouped)	FDR < 0.1 Metabolites (Detailed Grouped)
Cognitive_CogPos_vs_CogNeg	23	10	7	Lipid - Androgenic Steroids: androstenediol (3beta,17beta) disulfate (2)** Lipid - Corticosteroids: cortisone Lipid - Phosphatidylserine (PS): 1-stearoyl-2-arachidonoyl-GPS (18:0/20:4) Lipid - Pregnenolone Steroids: 21-hydroxypregnenolone disulfate**, 21-hydroxypregnenolone monosulfate (1)*, pregnenediol disulfate (C21H34O8S2)**, 17alpha-hydroxypregnenolone 3-sulfate*, pregnenetriol disulfate* Lipid - Progestin Steroids: 5alpha-pregnan-3beta,20alpha-diol disulfate* Nucleotide - Purine Metabolism, Adenine containing: adenosine 3',5'-cyclic monophosphate (cAMP)*	Lipid - Androgenic Steroids: androstenediol (3beta,17beta) disulfate (2)** Lipid - Pregnenolone Steroids: 21-hydroxypregnenolone disulfate**, 21-hydroxypregnenolone monosulfate (1)*, pregnenediol (C21H34O8), 17alpha-hydroxypregnenolone 3-sulfate* Lipid - Progestin Steroids: 5alpha-pregnan-3beta,20alpha-diol disulfate* Nucleotide - Purine Metabolism, Adenine containing: cyclic monophosphate (cAMP)*
Cognitive_CogPos_vs_Control	51	17	7	Carbohydrate - Glycolysis, Gluconeogenesis, and Pyruvate Metabolism: lactate Energy - Glycolysis, Gluconeogenesis, and Pyruvate Metabolism: Glycolytic_Ratio*	Energy - Glycolysis, Gluconeogenesis, and Pyruvate Metabolism: Glycolytic_Ratio* Lipid - Fatty Acid Metabolism:

				<u>Energy - TCA Cycle:</u> citrate <u>Lipid - Dihydrosphingomyelins:</u> myristoyl dihydrosphingomyelin (d18:0/14:0)* <u>Lipid - Fatty Acid Metabolism (Acyl Carnitine, Hydroxy):</u> (S)-3-hydroxybutyrylcarnitine* <u>Lipid - Fatty Acid Metabolism (Acyl Carnitine, Short Chain):</u> acetylcarnitine (C2) <u>Lipid - Lysophospholipid:</u> 2-myristoyl-GPC (14:0)** , 1-myristoyl-GPC (14:0) <u>Lipid - Phosphatidylcholine (PC):</u> phosphatidylcholine (14:0/14:0, 16:0/12:0)*, 1-myristoyl-2-palmitoyl-GPC (14:0/16:0)*, 1-oleoyl-2-docosaheptaenoyl-GPC (18:1/22:6)*, 1-palmitoyl-2-oleoyl-GPC (16:0/18:1), 1-palmitoyl-2-palmitoleoyl-GPC (16:0/16:1)*, 1-myristoyl-2-linoleoyl-GPC (14:0/18:2)* <u>Lipid - Primary Bile Acid Metabolism:</u> glyco-beta-muricholate** <u>Lipid - Progestin Steroids:</u> 5alpha-pregnan-3beta,20alpha-diol monosulfate (2)*, 5alpha-pregnan-3beta,20alpha-diol disulfate*	Carnitine, F hydroxybuty <u>Lipid - Lyso</u> 2-myristoyl- <u>Lipid -</u> <u>Phosphatidy</u> phosphatidy (14:0/14:0, 1 myristoyl-2- GPC (14:0/ <u>Lipid - Prog</u> 5alpha-preg 3beta,20alp monosulfate pregnan-3be diol disulfate
Subtypes_Cognitive_vs_Control	67	21	6	<u>Carbohydrate - Glycolysis, Gluconeogenesis, and Pyruvate Metabolism:</u> lactate <u>Energy - Glycolysis, Gluconeogenesis, and Pyruvate Metabolism:</u> Glycolytic_Ratio <u>Lipid - Fatty Acid Metabolism (Acyl Carnitine, Hydroxy):</u> (S)-3-hydroxybutyrylcarnitine, 3-hydroxyoctanoylcarnitine (1)	<u>Lipid - Long</u> <u>Polyunsatur</u> <u>Acid (n3 and</u> tetradecadi <u>Lipid - Lyso</u> 2-myristoyl- 1-myristoyl- <u>Lipid -</u> <u>Phosphatidy</u> 1-myristoyl-

			<u>Lipid - Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated):</u> myristoleoylcarnitine/physeteroylcarnitine (C14:1)* <u>Lipid - Fatty Acid Metabolism (Acyl Carnitine, Short Chain):</u> acetylcarnitine (C2) <u>Lipid - Long Chain Polyunsaturated Fatty Acid (n3 and n6):</u> tetradecadienoate (14:2)**, docosahexaenoate (DHA; 22:6n3) <u>Lipid - Lysophospholipid:</u> 2-myristoyl-GPC (14:0)***, 1-myristoyl-GPC (14:0)*, 1-palmitoleoyl-GPC (16:1)* <u>Lipid - Phosphatidylcholine (PC):</u> 1-myristoyl-2-palmitoyl-GPC (14:0/16:0)*, phosphatidylcholine (14:0/14:0, 16:0/12:0)*, phosphatidylcholine (15:0/18:1, 17:0/16:1, 16:0/17:1)*, 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*, 1-palmitoyl-2-oleoyl-GPC (16:0/18:1), 1-stearoyl-2-docosahexaenoyl-GPC (18:0/22:6), 1-margaroyl-2-oleoyl-GPC (17:0/18:1)* <u>Lipid - Primary Bile Acid Metabolism:</u> glycochenodeoxycholate <u>Lipid - Progestin Steroids:</u> 5alpha-pregnan-3beta,20alpha-diol disulfate*, 5alpha-pregnan-3beta,20alpha-diol monosulfate (2)	GPC (14:0/14:0), phosphatidylcholine (14:0/14:0, 16:0/12:0)* <u>Lipid - Progestin Steroids:</u> 5alpha-pregnan-3beta,20alpha-diol disulfate*
Subtypes_Depressive_vs_Control	46	20	9 <u>Amino Acid - Tryptophan Metabolism:</u> N-acetyltryptophan <u>Lipid - Androgenic Steroids:</u> androstenediol (3beta,17beta) disulfate (2)*, androstenediol (3beta,17beta) disulfate (1) <u>Lipid - Fatty Acid Metabolism (Acyl</u>	<u>Lipid - Androgenic Steroids:</u> androstenediol (3beta,17beta) disulfate (2)* <u>Lipid - Phosphatidylcholine:</u> 1-palmitoyl-docosahexaenoate

				<u>Carnitine, Polyunsaturated):</u> linoleoylcarnitine (C18:2)* <u>Lipid - Lysophospholipid:</u> 2-linoleoyl-GPC (18:2)*, 1-dihomo-linolenoyl-GPC (20:3n3 or n6)* <u>Lipid - Phosphatidylcholine (PC):</u> 1-palmitoyl-2-docosahexaenoyl-GPC (16:0/22:6)**, 1-stearoyl-2-docosahexaenoyl-GPC (18:0/22:6)**, 1-margaroyl-2-docosahexaenoyl-GPC (17:0/22:6)***, 1-pentadecanoyl-2-docosahexaenoyl-GPC (15:0/22:6)**, 1,2-dilinoleoyl-GPC (18:2/18:2), 1-oleoyl-2-docosahexaenoyl-GPC (18:1/22:6)* <u>Lipid - Phosphatidylethanolamine (PE):</u> 1-stearoyl-2-docosahexaenoyl-GPE (18:0/22:6)***, 1-palmitoyl-2-docosahexaenoyl-GPE (16:0/22:6)**, 1-oleoyl-2-docosahexaenoyl-GPE (18:1/22:6)* <u>Lipid - Pregnenolone Steroids:</u> pregnenetriol disulfate**, pregnenediol disulfate (C21H34O8S2)*, 21-hydroxypregnenolone disulfate <u>Lipid - Primary Bile Acid Metabolism:</u> glyco-beta-muricholate*** <u>Lipid - Sphingosines:</u> sphingosine 1-phosphate	(16:0/22:6)* docosahexa (18:0/22:6)* 2-docosahex (17:0/22:6)* pentadecan docosahexa (15:0/22:6)* <u>Lipid - Phosphatidylcholine (PC):</u> 1-stearoyl-2-docosahexaenoyl-GPC (18:0/22:6)***, 1-palmitoyl-2-docosahexaenoyl-GPC (16:0/22:6)** <u>Lipid - Pregnenolone Steroids:</u> pregnenediol disulfate** <u>Lipid - Primary Bile Acid Metabolism:</u> muricholate
Subtypes_MildPTSD_vs_Control	18	3	0	<u>Carbohydrate - Glycolysis, Gluconeogenesis, and Pyruvate Metabolism:</u> lactate <u>Lipid - Phosphatidylcholine (PC):</u> 1,2-dipalmitoyl-GPC (16:0/16:0) <u>Lipid - Secondary Bile Acid Metabolism:</u> deoxycholic acid glucuronide	
Subtypes_SeverePTSD_vs_Control	42	14	5	<u>Amino Acid - Glutamate Metabolism:</u>	<u>Energy - Glucose Metabolism:</u>

			glutamate <u>Amino Acid - Tryptophan Metabolism:</u> serotonin <u>Amino Acid - Urea cycle; Arginine and Proline Metabolism:</u> GABR <u>Carbohydrate - Glycolysis, Gluconeogenesis, and Pyruvate Metabolism:</u> lactate <u>Energy - Glycolysis, Gluconeogenesis, and Pyruvate Metabolism:</u> Glycolytic_Ratio* <u>Energy - TCA Cycle:</u> citrate <u>Lipid - Long Chain Polyunsaturated Fatty Acid (n3 and n6):</u> docosahexaenoate (DHA; 22:6n3) <u>Lipid - Phosphatidylcholine (PC):</u> 1-pentadecanoyl-2-docosahexaenoyl-GPC (15:0/22:6)***, 1-oleoyl-2-docosahexaenoyl-GPC (18:1/22:6)**, 1-margaroyl-2-docosahexaenoyl-GPC (17:0/22:6)**, 1-palmitoyl-2-(hydroxylinoleoyl)-GPC (16:0/18:2(OH))* <u>Lipid - Secondary Bile Acid Metabolism:</u> ursodeoxycholate <u>Lipid - Sphingolipid Synthesis:</u> sphinganine-1-phosphate <u>Lipid - Sphingosines:</u> sphingosine 1-phosphate**	Gluconeogenesis Pyruvate Metabolism Glycolytic_Ratio* Lipid - Phosphatidylcholine 1-pentadecanoyl-2-docosahexaenoyl-GPC (15:0/22:6)***, 1-oleoyl-2-docosahexaenoyl-GPC (18:1/22:6)***, 1-margaroyl-2-docosahexaenoyl-GPC (17:0/22:6)** Lipid - Sphingolipid Synthesis sphingosine
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